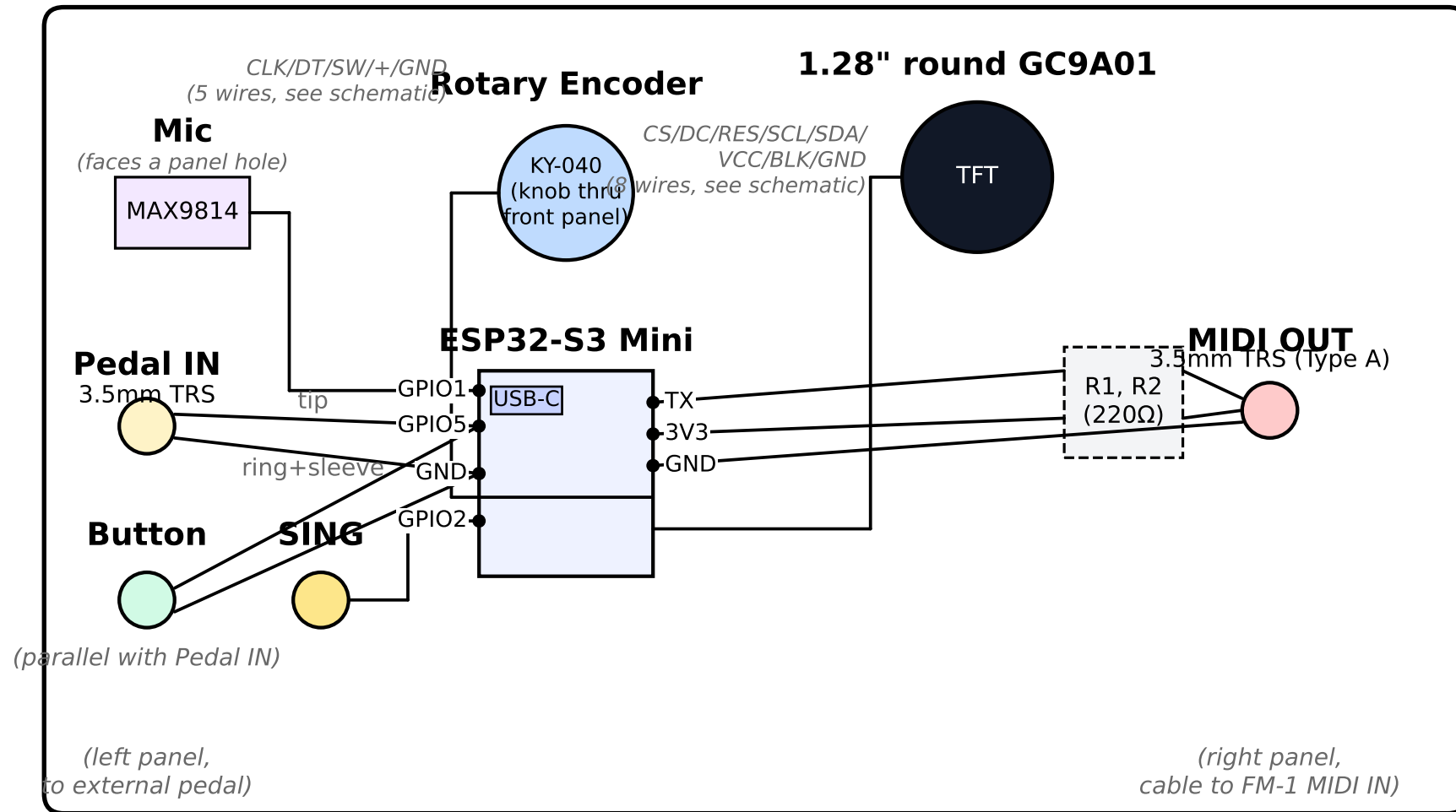


FM-1 Control Box (v2) — Layout / Assembly Diagram

Top-down view of enclosure: pedal jack, button, SING button (left panel), mic module (behind a front-panel hole), ESP32-S3 (center), MIDI out jack (right panel), encoder + round T

project box (top cover removed)



Notes:

- Resistors R1/R2 can live on a small offcut of perfboard, or be soldered directly in-line on the wire runs -- exact placement isn't critical
- See the schematic for the electrical connections (which wire goes through which resistor, which pin is which) -- this diagram is for physical placement only
- Button and Pedal IN jack are wired in parallel to the same GPIO5/GND nodes -- either one alone triggers sustain, so the pedal can be left unplugged
- Encoder knob and TFT window both need cutouts in the front panel -- mock up the panel layout before drilling, this diagram isn't to physical scale
- Leave slack on the USB-C cable path -- the board still needs to be reachable for re-flashing
- v1's pitch-strip input was dropped in this rebuild (see project README History) -- no strip or its wiring here